



MUHAMMAD TAHA

FULL-STACK AI ENGINEER

Full Stack AI Engineer specializing in end-to-end deployment, bridging advanced machine learning with scalable web architectures. Built edge-quantized vision and speech models running under 4 MB on Raspberry Pi 5, calibrated clinical prediction ensembles over 40GB+ of EHR data, and production Next.js platforms serving live US healthcare clients. Adept at full-stack systems design across low-latency API pipelines, dual-database architectures, and biometric verification suites.



taha-portfolio-website.vercel.app



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EDUCATION

Bachelor of Computer Engineering

National University of Science and Technology (NUST)

2022-2026

Relevant Coursework: Fundamentals of Programming, Object Oriented Programming, Data Structures & Algorithms, Software Engineering, Digital Image Processing, Artificial Intelligence & Decision Support System, Machine Learning, Computer Vision



PROFESSIONAL EXPERIENCE

Full Stack Engineer

A&T Nexus Solutions LLC | Remote

Aug 2025 - Present

- **Client Delivery & Ownership:** Sole engineer for 4 production Next.js platforms (healthcare/B2B), owning architecture and deployment across 290+ commits.
- **Security & Accessibility Baseline:** Standardized hardened defaults (CSP, X-Frame-Options, MIME-sniff headers), form honeypots, HTML escaping, and WCAG-compliant UIs.
- **Data & Integrations Layer:** Engineered JWT-secured admin APIs over Prisma/PostgreSQL and pooled MySQL. Brokered Vercel-Blob uploads and workflows (Resend, Sheets, WhatsApp).
- **Legacy Migration:** Authored 15+ Node scripts over SSH-tunneled MySQL to port WordPress to Next.js, automating HTML extraction, WebP conversion, and asset validation.
- **B2B Digital Transformation Platform:** Built flagship React 19/TS site with @react-three/fiber 3D scenes and Framer Motion storytelling, backed by validated serverless routes.

Computer Vision Intern

Jul 2025 - Sep 2025

TruID Technologies PVT. LTD. | NSTP, Islamabad, Pakistan

- **Biometric Anti-Spoofing:** Built a MobileNetV2 screen-replay detector. Applied CLAHE & unsharp masking over 17.8k images for 99.8% validation accuracy on face/hand spoofs.
- **ID Fraud Prevention:** Architected an FPN with CBAM attention. Used LAB-space CLAHE & edge-sharpening to isolate splice artifacts, overcoming 1:5 imbalance for 93.8% accuracy & 0.97 precision.
- **Signature Verification:** Designed a deep metric learning verifier via custom hard-negative mining & Triplet Loss. Mapped signatures to 256-D embeddings, yielding 92.7% accuracy across 5k pairs.
- **Model Optimization:** Built Tesla P100 GPU inference pipelines for all 4 vision models, profiling evaluation steps to guarantee a strict ≤ 31 ms batch latency for scalable KYC deployment.



TECHNICAL SKILLS

- **Languages & Core Tools:** Python, TypeScript, JavaScript, C++, SQL, MATLAB, Bash / Linux, Git
- **Deep Learning, LLMs & MLOps:** PyTorch, TensorFlow, Keras, Hugging Face, LLMs (Qwen, Ollama), RAG, LoRA Fine-Tuning, LightGBM, XGBoost, Optuna, SHAP
- **Vision, Audio & Edge AI:** OpenCV, Librosa, Whisper, YOLOv8, ResNet / MobileNet, 3DGS, Edge AI (TFLite INT8/NF4, ONNX), Raspberry Pi 5
- **Frontend Web Architecture:** Next.js (App Router), React, Tailwind CSS, Framer Motion, Vite, WCAG Accessibility
- **Backend, APIs & Cloud:** FastAPI, Flask, Node.js, REST APIs, SSE Token Streaming, PostgreSQL, MySQL, AWS EC2



PRACTICAL EXPERIENCE

Machine Learning and AI Systems

SERENITY: Smart Emotion Recognition & Neural Intervention Technology

- **Edge AI & Perception:** Trained ResNet-18 FER (93.0%) and CNN-BiLSTM SER (80.6%) on a balanced 53k-utterance corpus; INT8/NF4-quantized to 3.86 MB on Raspberry Pi 5.
- **Hybrid Systems Architecture:** Architected a privacy-first edge-cloud pipeline running Whisper STT locally, streaming metadata to an EC2 LLM over SSE at 8-12s latency.
- **LLM Safety & EHR Integration:** Built a rule-based router computing risk independently of the LLM, feeding a React 18 EHR dashboard with PHQ-9/GAD-7 tracking.

MedTraceAI: ED to ICU Clinical Deterioration Model

- **Data Engineering:** Built an out-of-core Polars pipeline streaming 37.1M vital rows from 40GB+ of MIMIC-IV into a hospital-agnostic OMOP/LOINC triplet schema.
- **Architecture & Tuning:** Fused a 192-feature XGBoost branch and Clinical Event Transformer into an isotonic-calibrated ensemble via 300-trial Optuna.
- **Clinical Impact:** Forecast ICU transfer 6 hours early at 0.892 AUROC, 0.667 PR-AUC and 0.051 Brier across 19,248 ED visits, with SHAP audit trails.

3D Environment Reconstruction from Multi-View Images

- **Classical CV Pipeline:** Engineered an SfM pipeline over 49 calibrated DTU views using SIFT, FLANN-RANSAC and DLT triangulation for a 30,500-pt cloud.
- **MVS Densification:** Plane-sweep Multi-View Stereo scoring 96 depth hypotheses/pixel via illumination-invariant NCC, cleaning 166,303 pts to 130k.
- **Neural Rendering:** Trained a 3D Gaussian Splatting model over 15,000 iterations with MVS initialization, reaching ~25 dB PSNR at 0.04 L1+SSIM loss.

Full-Stack Software Engineering

Client work delivered at A&T Nexus Solutions LLC.

Glorious Home Care: Multi-Region Care Platform

- **Architecture & Data:** Architected a 34-page platform via a typed local content layer. Used `generateStaticParams` to pre-render 25+ dynamic routes, eliminating database round-trips.
- **SEO & Security:** Programmed per-route canonicals and OG tags for regional SEO. Hardened delivery with strict Content-Security-Policy, X-Frame-Options DENY, and MIME headers.
- **Interface & APIs:** Engineered an `IntersectionObserver` scroll-spy directory, a `requestAnimationFrame` marquee, and Framer Motion reveals, backed by serverless Resend intake.

Benevolence Home Services: Zero-CMS Web App

- **Architecture & Performance:** Built a Next.js 16/React 19 app via a local data engine, using `generateStaticParams` to pre-render service routes and a blog, eliminating database latency.
- **SEO & Content Parsing:** Automated JSON-LD schemas and sitemaps. Engineered a safe, dependency-free parser to style legacy blog markup without vulnerable DOM injections.
- **Backend & Hardening:** Built serverless APIs integrating Resend and a Google Sheets CRM via JWT auth, hardened with honeypots, HTML escaping, and graceful degradation.

Full-Stack Dental Practice Platform & Headless CMS

- **Architecture & CMS:** Engineered a Next.js 14 site and admin CMS with drag-and-drop reordering. Secured by edge middleware, `httpOnly` JWTs, and multi-event inactivity auto-logout.
- **Dual-Database Backend:** Paired Prisma/PostgreSQL with a lazy MySQL pool for legacy data. Built a Vercel Blob upload pipeline, slashing update times by 90% (30+ to <2 mins).
- **Performance & APIs:** Achieved <100kB loads via SSR and automated cache revalidation. Built SMTP appointment routing and resilient client data fetching with exponential backoff.

LICENSES & CERTIFICATIONS

- Deep Learning Specialization
- Machine Learning Specialization
- AI for Everyone
- Introduction to Front-End Development
- CS50: Introduction to Programming with Python

DeepLearning.AI | Dec 2025

DeepLearning.AI | Jul 2025

DeepLearning.AI | Jun 2024

Meta | Oct 2023

Harvard's CS50 | Sep 2023